BGIN Block#12 Meeting Report

- BGIN is a multi-stakeholder discussion body / distributed think tank.

Layer2 Governance

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BGIN Block#12

Layer2 Governance

Date: Sunday, March 2, 2025

Location: Tokyo, Japan

Duration: Approx. 90 minutes

Summary

<u>"What is Layer 2 (L2)? What comes to your mind when you hear the term?"</u>

Participant Responses (via sticky notes):

- Rollups
- Faster transaction speed
- Lower transaction costs
- Scalability
- Application-specific chains
- Block space abundance

- The term Layer 2 is misunderstood and varies:
  - Technically: Specific implementations like Optimistic and ZK rollups.
  - Culturally: Associated with ideas like decentralization, accessibility, user empowerment, etc.

- Noted confusion in the industry: many distinct technologies (rollups, state channels, sidechains) are all referred to as "Layer 2."

- Suggested that this confusion is part of why the workshop is necessary.

- Key Takeaway/Framing:
  - A need to clarify what L2 is, both technically and from a governance perspective.
  - Importance of understanding how L2s interact with L1 governance.

<u>Ethereum's Goals & L1 Governance Perspective</u>

- Framing Quotes Introduced: Quote from an L2 developer (from a previous session): "L2 is a full-layer blockchain that allows fast and efficient transactions without damaging L1 governance."

- Clarifications on the Scope: The discussion will focus on Ethereum (not other L1s). Governance here means Ethereum Foundation's view, not DAOs or regulators. Regulation-related governance and DAO governance may be discussed later, but not

the current focus.

- Sticky Note Brainstorming Prompt: “What are Ethereum’s assets and goals?”

  Participants added notes including:

  - Self-sovereignty
  - Decentralization
  - Smart contract programmability
  - Proof of Stake (PoS)
  - Developer community empowerment
  - Pluralism (diverse views and approaches are welcomed)
  - Security and trust minimization
  - Credible neutrality

- Pluralism was emphasized as a key Ethereum value that contrasts with Bitcoin’s tighter, simpler governance. Ethereum’s culture supports diverse ideologies, communities, and use cases. The developer community’s empowerment is central to Ethereum’s ecosystem governance.

<u>Comparing Current Status of Ethereum Governance with Bitcoin</u>

- Key Axes for Comparison: Participants used sticky notes to compare Ethereum and Bitcoin on several dimensions:

  - Operating Organization
    - Ethereum has the Ethereum Foundation.
    - Bitcoin has no central organization, but relies on core developers.
  - Development Funding
    - Ethereum: Primarily through the Foundation and grants.
    - Bitcoin: Community donations and independent orgs.
  - Consensus Mechanism
    - Ethereum: Proof of Stake (PoS).
    - Bitcoin: Proof of Work (PoW).
  - Node Operation
    - Ethereum: More demanding, especially with validators.
    - Bitcoin: Easier for users to run full nodes.
  - Protocol Complexity
    - Ethereum: Highly complex (smart contracts, EVM).
    - Bitcoin: Simpler scripting and focus on money use case.
  - Monetary Policy
    - Bitcoin: Fixed supply (21M BTC).
    - Ethereum: Dynamic, supply can change (e.g., EIP-1559, Merge).
  - Governance Structure
    - Ethereum: Involves Foundation, devs, researchers, validators.

        - Bitcoin: Mostly developers and miners.

- Key Issues Raised: Decentralization is Relative: Ethereum claims to value decentralization, but critics argue it is increasingly centralized.
    - Key issue: Which layer or domain is centralized?
        - Governance layer?
        - Staking layer?
        - Dev ecosystem?

- Need for Specificity:
    - Centralization varies by:
        - Protocol governance
        - Staking and validator concentration
        - App and infra ecosystem
    - Example: Ethereum has a large dev ecosystem, which is less centralized, but validator nodes may be concentrated.

- Pointed out that Ethereum governance must be analyzed across multiple layers:
    - Protocol devs
    - Foundation influence
    - Client teams
    - Validators and staking services
    - App developers and community contributors

- Noted that Bitcoin's mining centralization (e.g., large mining pools) also challenges its decentralization narrative.

- Emphasized that centralization/decentralization is a spectrum, not a binary:
    - Even the Federal Reserve is centralized but has branches.
    - Ethereum's governance might be *decentralized in name but not in practice* (example: whale voting in DAOs).

- Notable Quote: "Centralization is not black or white. It depends on what part of the stack you're looking at—staking, devs, infra, or governance itself."

<u>Can L2s Undermine Ethereum's Governance?</u>

- Referenced a quote from a past session: "L2 is a full VM blockchain that allows fast and efficient transactions, and does not damage L1 governance." The group discussed what it means for L2s to respect Ethereum's L1 governance and ethos

- Key question raised: What conditions must an L2 meet to avoid undermining Ethereum's governance, assuming Ethereum governance aligns with its assets and values?

- Key Concepts Raised: L2s should preserve Ethereum's core values, such as:

    - Censorship resistance
    - Open-source transparency
    - Permissionlessness
    - User sovereignty
    - Decentralized governance

- Ethereum cannot prevent anyone from launching an L2, since it's permissionless.
    - Ethereum cannot censor rollups or prevent L2s from posting commitments on-chain.
    - So it becomes a cultural and security question, not a formal enforcement issue.

- Two Perspectives on Governance Risk:

    - Technical Undermining:

        - Could occur if an L2 grows large enough to control or influence L1 governance (e.g., validator collusion, overwhelming stake influence)
        - Example: Risk of restaking via EigenLayer concentrating influence.

    - Cultural/Normative Undermining:

        - Happens when L2s ignore Ethereum's cultural values (e.g., centralization, censorship).
        - Even if technically sound, an L2 could undermine Ethereum's identity and community ethos.

- Should L2s be required to be censorship-resistant like Ethereum?

    - Even if an L2 has censorship, as long as users have the ability to withdraw back to L1, then Ethereum's guarantees are preserved. Others argued: It's better if L2s also inherit censorship resistance natively, or else risks remain.

<u>Attack Vectors in Layer 2 Architecture</u>

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1. Smart Contract Governance (L1 Bridge Contract):

- The L1 smart contract that locks ETH is often administered by the L2 team or foundation.
- This is a single point of failure.
- Risk: if this contract is upgradable or centrally governed, it could be exploited or censored.

2. Sequencer Centralization:

- Sequencers bundle and submit L2 transaction data to L1.

- Many L2s (e.g. Arbitrum, Optimism) use centralized sequencers for performance.
- Risks:
    - Censorship: sequencer may refuse to include a user's transaction.
    - Reordering or front-running.
    - Downtime or malicious submission of data to L1.

3. Data Availability Layer:

- Some L2s post data off-chain to save gas. If this off-chain data is lost or manipulated, users may lose access to prove their balances or exit.

- L2s are developing data availability layers (e.g., Celestia) with new economic security (by issuing tokens, staking, slashing). But these are new trust assumptions. E.g., "We'll punish bad behavior with token slashing." Very few powerful nodes with real computational power might dominate.

- There's a trade-off between performance (scalability) and security: Posting all data on-chain = most secure, least scalable. Off-chain data = scalable, but increases trust and attack surfaces.

- EIP-4844 was brought up: Enables blobs to post L2 data to L1 cheaply (intermediate solution). A step to avoid off-chain storage while maintaining efficiency.

- Off-chain DA layers include systems like Celestia, Avail, and EigenDA. These systems use their own tokens and staking to create economic security. Participants in DA layers stake tokens and get slashed for dishonest behavior (e.g., hiding data). Each L2 ends up creating its own token ecosystem and validator set. This raises questions about added complexity and whether it actually improves decentralization.

- ZK Rollups are **more secure and faster**, but not as scalable today. Optimistic Rollups are **more widely deployed** due to simplicity and EVM compatibility. Many teams are exploring **hybrid approaches** to combine benefits of both.

- The group agreed: All L2s share similar architecture, but differ in their trade-offs across decentralization, speed, and security.

<u>Role of Users & Use Case-Specific L2 Governance</u>

- Users are often unaware or indifferent to the degree of decentralization of each L2. The most popular L2s today are highly centralized but offer low fees and fast transactions. From a user perspective, low cost and speed often outweigh decentralization concerns.

- There is a fundamental governance issue: centralization is incentivized because it's more efficient. This could ultimately harm Ethereum's long-term vision and values if L2s

diverge too far. The Ethereum Foundation doesn't directly control L2s and can't enforce governance standards.

- Some participants suggest introducing a threshold or classification system for L2s based on decentralization level. Enterprises may avoid centralized L2s due to security and compliance needs.
- The conversation shifted toward how to educate users and create awareness around L2 risks. There's a need for metrics or dashboards that compare L2s not just on performance but also on decentralization and governance.
    - Participants proposed focusing on L2 attributes to better understand their strengths and ideal use cases. Example: Optimism is aligned with public goods funding due to its governance model and incentives.
    - Each L2 could find a specialized niche in the broader Ethereum ecosystem (e.g., trading, gaming, social, etc.). This niche specialization allows for a more modular and pluralistic Ethereum.
    - There's potential value in researching which L2 architectures are best suited for specific use cases.
- Participants agreed to shift focus of the working group paper from EF's view to user perspective—examining how users assess and choose L2s based on tradeoffs like cost vs. decentralization.

<u>Next Steps</u>

- We will have a reflection meeting. An L2-specific working group session will be scheduled in the coming weeks. IKP working group will rotate its three topics (including L2 governance) biweekly over the next two months.